

$$V = \text{span} \left\{ \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \\ 2 \end{pmatrix} \right\}$$

$$f(v_1) = \begin{pmatrix} 4 \\ 1 \\ 2 \\ 5 \end{pmatrix} \quad f(v_2) = \begin{pmatrix} 2 \\ 2 \\ -2 \\ -2 \end{pmatrix}$$

$$\text{base } B = (v_1, v_2)$$

$$f(v_1) = v_1 + 2v_2$$

$$f(v_2) = 2v_1 - 2v_2$$

$$\text{matrix } f: A = \begin{pmatrix} 1 & 2 \\ 2 & -2 \end{pmatrix}$$

$$\det(A - \lambda I_2) = (1-\lambda)(-2-\lambda) - 4 = \lambda^2 + \lambda - 6$$

$$\lambda_1 = 2$$

$$\lambda_2 = -3$$

$$\lambda_1 = 2: \begin{pmatrix} -1 & 2 \\ 2 & -4 \end{pmatrix} \sim \begin{pmatrix} 1 & -2 \\ 0 & 0 \end{pmatrix} \quad \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

$$\lambda_2 = -3: \begin{pmatrix} 4 & 2 \\ 2 & 1 \end{pmatrix} \sim \begin{pmatrix} 2 & 1 \\ 0 & 0 \end{pmatrix} \quad \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

$$\text{base } C = \left(\begin{pmatrix} 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -2 \end{pmatrix} \right)$$

2 v. rekomb

$$[f_A]_C^C = \text{diag}(2, -3)$$

$$x = \begin{pmatrix} 5 \\ 1 \\ 3 \\ 2 \end{pmatrix}$$

$$[x]_B = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

$$\left(\begin{array}{cc|c} 2 & 1 & 1 \\ 1 & -2 & 3 \end{array} \right)$$

$$[x]_C = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$\left\{ (P_A)^n \right\}_C^C = \left(\left\{ P_A \right\}_C^C \right)^n = \begin{pmatrix} 2^n & 0 \\ 0 & (-3)^n \end{pmatrix}$$

$$x_n := P^n \begin{pmatrix} 5 \\ 1 \\ 3 \\ 7 \end{pmatrix} :$$

$$\left\{ x_n \right\}_C = \begin{pmatrix} 2^n & 0 \\ 0 & (-3)^n \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 2^n \\ -(-3)^n \end{pmatrix}$$

$$\left\{ x_n \right\}_B = 2^n \cdot \begin{pmatrix} 2 \\ 1 \end{pmatrix} - (-3)^n \begin{pmatrix} 1 \\ -2 \end{pmatrix} = \begin{pmatrix} 2^{n+1} - (-3)^n \\ 2^n + 2(-3)^n \end{pmatrix}$$

$$x_n = \begin{pmatrix} 2^{n+1} - (-3)^n \\ 2^n + 2(-3)^n \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix} + \begin{pmatrix} 2^n + 2(-3)^n \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \\ 1 \\ 2 \end{pmatrix} =$$

$$= \begin{pmatrix} 5 \cdot 2^n \\ 2^{n+1} - (-3)^n \\ 2^n + 2(-3)^n \\ 2^{n+2} + 3(-3)^n \end{pmatrix}$$